Name:

Class:

UNIT EXAM WITH ANSWERS

Unit 4 Revolutions in modern physics

Time permitted: 70 minutes

	Section	Number of questions	Marks available
A	Multiple choice	30	30
B	Short answer	10	40
	Total		**70**

Scale:

A+	66–70	A	60–65	B	50–59	C	40–49	D	35–39	E	21–34	UG	0–20

For all questions:

- The speed of light is 2.998×10^{8} m s^{-1}
- The charge on an electron is -1.602×10^{-19} C
- The Planck constant, $h = 6.626 \times 10^{-34}$ J s $= 4.14 \times 10^{-15}$ eV s
- The Rydberg constant, $R = 1.097 \times 10^{-7}$ m^{-1}
- The charge on an electron is -1.602×10^{-19} C
- The mass of an electron is 9.11×10^{-31} kg
- The mass of a proton is 1.67×10^{-27} kg
- The Hubble constant, $H = 2.2 \times 10^{-2}$ m s^{-1} ly^{-1}
- Wein's constant, $b = 2.898 \times 10^{-3}$

Table 1 Properties of quarks

Name	Symbol	Spin	Charge	Strangeness	Charm	Bottomness	Topness
Up	u	$\frac{1}{2}$	$+\frac{2}{3}e$	0	0	0	0
Down	d	$\frac{1}{2}$	$-\frac{1}{3}e$	0	0	0	0
Strange	s	$\frac{1}{2}$	$-\frac{1}{3}e$	-1	0	0	0
Charmed	c	$\frac{1}{2}$	$+\frac{2}{3}e$	0	$+1$	0	0
Bottom	b	$\frac{1}{2}$	$\frac{1}{3}e$	0	0	$+1$	0
Top	t	$\frac{1}{2}$	$+\frac{2}{3}e$	0	0	0	$+1$

Table 2 Quark composition of mesons

		Antiquarks									
		$\bar{b}$		$\bar{c}$		$\bar{s}$		$\bar{d}$		$\bar{u}$	
Quarks	b	γ	$(\bar{b}b)$	B_c^-	$(\bar{c}b)$	$\bar{B}_s$	$(\bar{s}b)$	$\bar{B}_{d0}$	$(\bar{d}b)$	B^-	$(\bar{u}b)$
	c	B_c^+	$(\bar{b}c)$	J/ψ	$(\bar{c}c)$	D_s^+	$(\bar{s}c)$	D^+	$(\bar{d}c)$	D^0	$(\bar{u}c)$
	s	B_s^0	$(\bar{b}s)$	D_s^-	$(\bar{c}s)$	φ	$(\bar{s}s)$	$\bar{K}_0$	$(\bar{d}s)$	K^-	$(\bar{u}s)$
	d	B_d^0	$(\bar{b}d)$	D^-	$(\bar{c}d)$	K^0	$(\bar{s}d)$	π^0	$(\bar{d}d)$	π^-	$(\bar{u}d)$
	u	B^+	$(\bar{b}u)$	$\bar{D}_0$	$(\bar{c}u)$	K^+	$(\bar{s}u)$	π^+	$(\bar{d}u)$	π^0	$(\bar{u}u)$

Table 3 Quark composition of several baryons

Particle	Quark composition
p	uud
n	udd
Λ^o	uds
Σ^+	uus
Σ_c^{++}	uuc
Σ_b^+	uub
Σ_t^{++}	uut
Σ^o	uds
Σ^-	dds
Δ^{++}	uuu
Δ^+	uud
Δ^o	udd
Δ^-	ddd
Ξ^o	uss
Ξ^-	dss
Ω^-	sss

Section A Multiple choice (30 marks)

Section A consists of 30 questions, each worth one mark. Each question has only one correct answer. Circle the correct answer. Attempt all questions. Marks will not be deducted for incorrect answers. You are advised to spend no more than 30 minutes on this section.

1 Which of the following best explains the nature of light?

A Light is a form of wave motion with an unknown aether as the medium.

B Light is a form of wave motion with no medium needed.

C Light is made up of miniscule particles called photons.

D *Light is neither a wave nor a series of particles but can be modelled as both of these.*

2 How would you know you were in an inertial frame of reference?

A If you dropped something, it would fall to the ground.

Ⓑ *You would be 'weightless', and so would everything else.*

C The speed of light would be constant.

D There would be no way you could tell.

3 A rocket is travelling at two-thirds the speed of light away from Earth. It flashes a signal back to Earth. At what speed is the signal travelling from the rocket to Earth?

A One-third the speed of light

B Two-thirds the speed of light

Ⓒ *The speed of light*

D Four-thirds the speed of light

4 A man in a very fast train, travelling at four-fifths the speed of light, hits a button sending a radio signal once every 0.6 s. At what frequency will another man, at rest, receive the signals?

A One signal every 0.3 s

Ⓑ *One signal every 0.36 s*

C One signal every 0.6 s

D One signal every 1.0 s

5 Barnard's star is 5.96 light-years from Earth. How far would this distance be for an astronaut travelling at $2.880 \times 10^{8}\ \mathrm{m\ s^{-1}}$?

A 0.23 light-years

Ⓑ *1.66 light-years*

C 30.0 light-years

D 151 light-years

6 The largest nuclear bomb ever detonated was the Tsar Bomba, releasing 2.1×10^{17} J of destructive energy. How much mass was converted to energy in this explosion?

A 0.25 g

B 14 g

C 870 g

Ⓓ *2.3 kg*

7 Which of the following black bodies would be hottest in temperature?

A It emits little or no light, but you can feel the heat.

B It is glowing a dull red.

Ⓒ *It is bright white, almost blueish.*

D It is glowing orange.

8 Why is the Sun considered a black body when it is clearly not black?

(A) *It absorbs nearly all of any light that it receives.*

B It does not absorb any light at all.

C It emits light across the whole spectrum.

D It is an extremely hot body.

9 What is the peak wavelength of light from slag being poured out of a blast furnace at 2600 K?

A 84 mm

(B) *1.1 μm*

C 3.4 μm

D 15.3 μm

10 Photons from most gamma rays emitted by technetium-99m contain 140.5 keV of energy. What is the frequency of this radiation?

A 5.82×10^{10} Hz

B 7.29×10^{15} Hz

C 4.01×10^{17} Hz

(D) *3.39×10^{19} Hz*

11 What is the minimum frequency required to emit electrons from sodium metal, with a work function of 3.94×10^{-19} J?

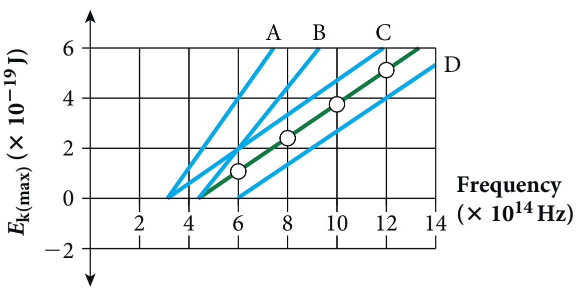
(A) *5.95×10^{14} Hz*

B 2.43×10^{15} Hz

C 8.40×10^{15} Hz

D 1.31×10^{17} Hz

12 The green graph below represents the maximum kinetic energy of ejected electrons against frequency of light on a particular metal. Which line would represent the corresponding graph for a metal that releases electrons more readily?



A Line A

B Line B

(C) *Line C*

D Line D

13 When light from a sodium lamp is viewed through a spectroscope, it is seen to be made up mainly of various lines. These lines are caused by:

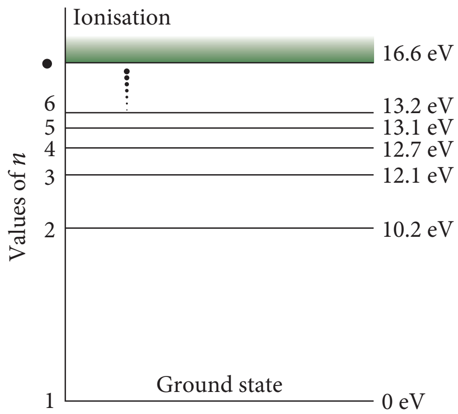
A *electrons dropping back to lower energy levels.*

B electrons absorbing energy and jumping to higher energy levels.

C sodium reacting with chemicals in the lamp.

D electrons being ejected from the sodium atoms.

14 The diagram below shows energy levels for the hydrogen atom.



If a photon of energy 1.0 eV was emitted from a hydrogen atom, what would happen?

A An electron from the second energy level could drop part way to the first energy level.

B An energy from the third energy level would shift to the fifth energy level.

C *An energy from the fifth energy level would shift to the third energy level.*

D An energy from the sixth energy level would shift to the third energy level.

15 Find the wavelength of light ejected as an electron drops from energy level 5 to 2.

A 366 nm

B *428 nm*

C 549 nm

D 731 nm

16 Find the de Broglie wavelength for a bullet of mass 3.2 g travelling at 1000 m s^{-1}.

A *2.1×10^{-34} m*

B 5.2×10^{-34} m

C 6.6×10^{-34} m

D 4.8×10^{-35} m

17 Calculate the fourth longest wavelength of an electron in an orbit of 68.0 nm.

A 53.4 nm

Ⓑ *107 nm*

C 214 nm

D 427 nm

18 Which of the following is true for a moving electron?

A It moves forward but also up and down like a particle in a wave.

B If you know its initial velocity and all conditions, you can theoretically predict its exact position at any time.

C It can only move with discrete amounts of kinetic energy.

Ⓓ *It can go through two or more narrow slits at the same time.*

19 Why might you suspect the neutron not to be a fundamental particle?

A It has no charge.

Ⓑ *It has a magnetic moment.*

C It interacts with the four forces.

D It is too large to be a fundamental particle.

20 Convert the mass of an electron into MeV/c^2.

Ⓐ *0.511* MeV/c^2

B 1.06 MeV/c^2

C 12.5 MeV/c^2

D 36.9 MeV/c^2

21 Why are particle accelerators used to create particles of high mass?

A Time passes more slowly on high speed particles, increasing their half-life.

B When a fast-moving particle collides, anything can happen.

Ⓒ *The increased energy enables particles of more mass to be created.*

D $E = mc^2$ means you need a lot of energy to create a small mass.

22 A pion is a:

Ⓐ *field particle.*

B lepton.

C baryon.

D kind of force.

23 Why can a neutron not simply decay to a proton and an electron without emitting an anti-neutrino?

A Baryon number would not be conserved.

B *Lepton number would not be conserved.*

C Charge would not be conserved.

D Spin would not be conserved.

24 Which force is mediated by photons?

A *Electromagnetic*

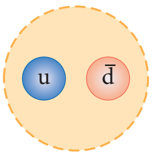
B Gravitational

C Weak

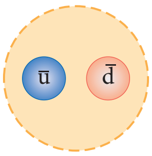
D Strong nuclear

25 Which of the following could be an antibaryon?

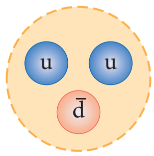
A



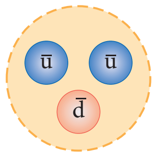
B



C



D



26 Using Table 1, determine the charge of a particle made up of two up quarks and a charmed quark.

A 0

B 1

C *2*

D −1

27 Describe a possible process for a π^- meson mediating the strong nuclear force.

A *Neutrons emit gluons that create a u quark and a $\bar{u}$ quark.*

B Protons emit gluons that create a u quark and a $\bar{u}$ quark.

C Neutrons emit gluons that create a d quark and a $\bar{d}$ quark.

D Protons emit gluons that create a d quark and a $\bar{d}$ quark.

28 Using red shift, a star's velocity relative to Earth has been calculated at 660 m s^{-1}. Use Hubble's law to estimate its distance away from us.

A 1500 light-years

B 3000 light-years

C 15 000 light-years

D *30 000 light-years*

29 Which of the following particles would have existed for the first few moments after the Big Bang?

Ⓐ *Gluons*

B Electrons

C Neutrons

D Neutrinos

30 What is dark energy?

A Energy emitted from black bodies as electromagnetic radiation

B Background radiation in the Universe, mainly of microwave frequencies

Ⓒ *Hypothetical energy that repels matter, increasing the expansion of the Universe*

D The missing matter in the Universe.

Section B Short answer (40 marks)

Section B consists of 10 questions. Write your answers in the spaces provided. You are advised to spend 40 minutes on this section.

1 A train passes a signal box at 20.0 m s^{-1}. At the same time, a person in a carriage walks backwards at 3 m s^{-1}. After 6.0 s, what is the person's displacement with respect to:

a the signal box?

Answer: Relative to the signal box, velocity = 15.0 m s^{-1} − 3.0 m s^{-1} = 12.5 m s^{-1}. (1 mark)
After 6.0 s, displacement is 12.0 m s^{-1} × 6.0 s = 72 m (1 mark)

b the train?

Answer: Relative to the train, the person is moving at −3.0 m s^{-1}. (1 mark)
After 6.0 s, displacement is −3.0 m s^{-1} × 6.0 s = −18 m (1 mark)
(= 4 marks total)

2 A proton of mass 1.672 × 10^{-27} kg is accelerated from rest by a potential difference of 400 MV. Find the final speed of the proton.

Answer: $E = qV$ and $\Delta E_k = (\gamma - 1)m_0c^2$ (1 mark)

$$(\gamma - 1)m_0c^2 = qV \Rightarrow \gamma = \frac{qV}{m_0c^2} + 1$$ (1 mark)

$$\gamma = \frac{1.602 \times 10^{-19}\ \text{C} \times 4.0 \times 10^{8}\ \text{V}}{1.672 \times 10^{-27}\ \text{kg} \times (2.998 \times 10^{8}\ \text{m s}^{-1})^2} + 1 = 1.426$$ (1 mark)

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 1.426 \Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{1.426^2} = 0.4915$$

$v^2 = c^2 \times (1 - 0.4915) = 0.5085\,c^2$
$v = 2.138 \times 10^8$ m s^{-1}. (1 mark)
(= 4 marks total)

3 In an experiment, a viewing screen is placed 3.00 m away from a pair of slits that are 0.25 mm apart. Light of frequency 5.54×10^{14} Hz is incident on the slits. What should be the positions of the first three bright bands?

Answer: The placing of the bright spots is $y = \frac{Lm\lambda}{d}$ where $m = 0, 1, 2, \ldots$ (1 mark)

$$c = f\lambda \Rightarrow \lambda = \frac{c}{f} = \frac{2.998 \times 10^{8}\ \text{m s}^{-1}}{5.54 \times 10^{14}\ \text{s}^{-1}} = 5.41 \times 10^{-7}\ \text{m}$$

(1 mark)

The placing of the bright spots is $y = \frac{Lm\lambda}{d}$ where $m = 0, 1, 2, \ldots$

$m = 0$ gives the first position $= 0$ m

$m = 1$ gives the second position $= \frac{3.00\ \text{m} \times 1 \times 5.41 \times 10^{-7}\ \text{m}}{2.5 \times 10^{-4}\ \text{m}} = 6.49 \times 10^{-3}$ m or 6.5 mm

$m = 2$ gives the third position $= 1.30 \times 10^{-2}$ m or 13.0 mm (1 mark)

(= 3 marks total)

4 The work function for platinum is 6.35 eV.

a Calculate the cut-off frequency for platinum.

Answer: $W = hf_0 \Rightarrow f_0 = \frac{W}{h}$ (1 mark)

$f_0 = \frac{6.35\ \text{eV}}{4.14 \times 10^{-15}\ \text{eV s}} = 1.53 \times 10^{15}$ Hz (1 mark)

b Calculate the maximum kinetic energy of electrons emitted if light of wavelength 40 nm is incident on a platinum surface.

Answer: $KE_{max} = \text{hf} - \text{W}$ and $\text{f} = \frac{c}{\lambda} \Rightarrow KE_{max} = \frac{hc}{\lambda} - W$ (1 mark)

$$KE_{max} = \frac{4.14 \times 10^{-15}\ \text{eV s} \times 2.998 \times 10^{8}\ \text{m s}^{-1}}{4.0 \times 10^{-8}\ \text{m}} - 6.35\ \text{eV}$$

$KE_{max} = 31.0\ \text{eV} - 6.35\ \text{eV} = 24.7\ \text{eV}$ (one decimal place accuracy) (1 mark)

c What is the minimum kinetic energy of electrons emitted?

Answer: 0 eV (1 mark)

(= 5 marks total)

5 A beam of electrons is incident on a metal containing layers of atoms that are known to be spaced 8.40×10^{-11} m apart. A diffraction pattern is produced with the primary region of constructive interference at an angle of 23.4°.

a Find the wavelength of the electrons.

Answer: $m\lambda = 2d \sin\theta \Rightarrow 1 \times \lambda = 2 \times 8.40 \times 10^{-11}\ \text{m} \times \sin 23.4° \Rightarrow \lambda = 6.67 \times 10^{-11}$ m

Optional alternatives: 0.0667 nm or 66.7 pm (1 mark)

b Find the angles for the next region of destructive interference and the next region of constructive interference.

Answer: Next constructive region: $m\lambda = 2d \sin \theta \Rightarrow \sin \theta = \dfrac{m\lambda}{2d}$

$m = 2 \Rightarrow \sin \theta = \dfrac{2 \times 6.67 \times 10^{-11}\ \text{m}}{2 \times 8.40 \times 10^{-11}\ \text{m}} = 0.794\ldots \Rightarrow \theta = 52.6°$ (1 mark)

Next destructive region: $2d \sin \theta = (m + \dfrac{1}{2})\ \lambda \Rightarrow \sin \theta = \dfrac{(m + 0.5)\lambda}{2d}$, $m = 1$ (1 mark)

$\sin \theta = \dfrac{1.5 \times 6.67 \times 10^{-11}\ \text{m}}{2 \times 8.40 \times 10^{-11}\ \text{m}} = 0.595\ldots \Rightarrow \theta = 36.6°$ (1 mark)

(= 4 marks total)

6 A high energy photon of wavelength 2.100 nm collides with a stationary electron and then returns along its original path.

a Find the kinetic energy and the magnitude of the momentum of the photon.

Answer: $E = hf = \dfrac{hc}{\lambda} = \dfrac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m s}^{-1}}{2.100 \times 10^{-9}\ \text{m}} = 9.459 \times 10^{-17}\ \text{J}$ (1 mark)

$E = hf = \dfrac{hc}{\lambda}$ and $p = \dfrac{E}{c} \Rightarrow p = \dfrac{h}{\lambda}$ (Students may just write $p = \dfrac{h}{\lambda}$.)

$p = \dfrac{6.626 \times 10^{-34}\ \text{J s}}{2.100 \times 10^{-9}\ \text{m}} = 3.155 \times 10^{-25}\ \text{N m s}^{-1}$. (1 mark)

b Write the equations that must be solved to find the velocity v of the electron and the wavelength λ of the photon after collision. Use m and h to stand for the (known) mass of the electron and Planck's constant.

Answer: Use conservation of momentum and energy, remembering that momentum of the photon will change direction.

Momentum: $mv - \dfrac{h}{\lambda} = 3.155 \times 10^{-25}\ \text{N m s}^{-1}$ (1 mark)

Energy (kinetic): $\dfrac{1}{2}mv^2 + \dfrac{hc}{\lambda} = 9.459 \times 10^{-17}\ \text{J}$ (1 mark)

Note: Students could solve these using a CAS calculator, but this would take time and no physics knowledge is required.

(= 4 marks total)

7 A tau lepton decays into an electron, a tau neutrino and an electron antineutrino.

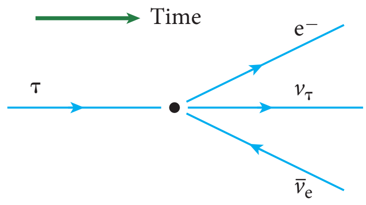
$\tau \rightarrow e^{-} + \nu_{\tau} + \overline{\nu}_{e}$

a Deduce the charge on a tau lepton. Reasoning must be shown.

Answer: Charge must be conserved.
Total charge on right side = −1 + 0 + 0 = −1
Therefore the tau lepton must have charge = −1. (1 mark)

b Draw this on a reaction diagram.

Answer:



Shape (1 mark)

Arrows (1 mark)

c Show that the lepton number is conserved.

Answer: Two lepton numbers must be considered.

Tau: Left side: 1 Right side: $0 + 1 + 0 = 1$ (1 mark)

Electron: Left side: 0 Right side: $1 + 0 + -1 = 0$ (1 mark)

d Write the time-reversal reaction. How likely is this to happen? Explain your answer.

Answer: $e^- + \nu_\tau + \bar{\nu}_e \rightarrow \tau$ (1 mark)

Very unlikely as three short-lived particles must come together. (1 mark)

e Apply crossing symmetry to the electron.

Answer: $\tau + e^+ \rightarrow \nu_\tau + \bar{\nu}_e$ (1 mark)

(= 8 marks total)

8 A baryon contains two down quarks and a strange quark.

a Calculate its charge, baryon number, strangeness and charm.

Answer: Charge $= -\frac{1}{3}e - \frac{1}{3}e - \frac{1}{3}e = -e$ ($\frac{1}{2}$ mark)

Baryon number $= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = +1$ (for any baryon) ($\frac{1}{2}$ mark)

Strangeness $= 0 + 0 + -1 = -1$ ($\frac{1}{2}$ mark)

Charm $= 0 + 0 + 0 = 0$ ($\frac{1}{2}$ mark)

b Calculate these same quantities for its antiparticle.

Answer: These quantities will all be reversed, so charge $= +e$, baryon number $= -1$, strangeness $= 1$ and charm $= 0$. (1 mark)

(= 3 marks total)

9 When a proton meets an antiproton, they do not annihilate each other like an electron and a positron. Instead, mesons are formed.

a Deduce which possible mesons could be formed, including the quarks and names of each meson.

Answer: A proton is uud and an antiproton is $\bar{u}\,\bar{u}\,\bar{d}$.

(1 mark)

You could get $u\bar{u}$, $d\bar{d}$, $u\bar{d}$ or $d\bar{u}$.

These are π^0, π^0, π^+ and π^- mesons respectively. (1 mark)

b What will be produced when a proton meets an antineutron, assuming that every quark joins with its corresponding antiquark?

Answer: A proton is uud and an antineutron is $\bar{u}\,\bar{d}\,\bar{d}$.

You would get $u\bar{u}$, $d\bar{d}$ and $u\bar{d}$

These are π^0, π^0 and π^+ mesons respectively.

(1 mark)

(= 3 marks total)

10 Explain what dark matter is and why it has been proposed.

Answer: Dark matter neither reflects nor radiates energy, hence it is dark. (1 mark)

Observations by astronomers concerning the velocity of objects indicate there is more matter in galaxies and the Universe in general than that calculated from the objects that we can see (mainly stars).

(1 mark)

(= 2 marks total)